# 6.8 Integration Using Trigonometric Substitution

Name _____ Period ____ Date _____

AP Calculus 6.8 · Student sheet

**I can…**

- choose the right trigonometric substitution.
- integrate and use a right triangle to return to x.
- evaluate a definite integral with a trigonometric substitution.

**Vocabulary** · fill in the missing word

**Trigonometric substitution** — _____ x with a sin θ, a tan θ, or a sec θ to remove a square root

**Reference triangle** — a right triangle labeled from the _____, used to write the answer in terms of x

## Part 1 · Triangle picture

x = 3 sin θ, so sin θ = x/3.

| Side or ratio | In terms of x |
|---|---|
| Opposite | |
| Hypotenuse | |
| Adjacent | |
| cos θ | |
| tan θ | |

## Part 2 · Notes and examples (fill in each note, then work the example)

**$\sqrt{(a^2 - x^2)}$:** let x = a _____ θ.

### Example 1

Find $\int dx/\sqrt{9 - x^2}$.

**$\sqrt{(x^2 + a^2)}$ or $x^2 + a^2$:** let x = a _____ θ.

### Example 2

Find $\int dx/(x^2 + 4)$.

**$\sqrt{(x^2 - a^2)}$:** let x = a _____ θ.

### Example 3

Which substitution should be used for an integral containing $\sqrt{x^2 - 16}$?

**Back to x:** use a right _____ labeled from the substitution (or change the limits).

### Example 4

Evaluate $\int_0^2 \sqrt{4 - x^2}\ dx$.

## Part 3 · Partner practice

### Partner 1

Find $\int dx/(x^2 + 25)$.

### Partner 2

(AP® Practice) $\int_0^3 \sqrt{9 - x^2}\ dx =$

(A) $\frac{9\pi}{4}$ (B) $\frac{9\pi}{2}$ (C) $3\pi$ (D) $9$